

Introduction to Economics

Class 6 — Equilibrium, shifts and prices as signals

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slide 1 of 24

Slide 1 Title

Class 6 puts Monday's two curves on one picture. The first hour: where they cross, what happens when the price sits above or below the crossing, and how the price does the work that no planner does. Then what happens to the crossing when a curve moves, in three steps that return in every problem of this kind from now on. The thread through the class is a real price: olive oil at origin, 3.60 euros a kilo in July 2022 and 9.20 in January 2024.

Quiz 3

12 minutes · pens down at 10:12 · questions on both sides

Circle the letter of your answer in the block at the top.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

quiz page, not numbered

Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letter of your answer in the block at the top of the sheet, write your name on the block, tear the strip off along the dashed line and hand in only the strip.

Plan for today

10:00	Quiz 3
10:13	Monday, one more look
10:15	Olive oil: what happened?
10:18	Where the two curves cross
10:22	Above and below the crossing
10:35	Prices as signals
10:40	When a curve moves: three steps
10:50	Break
11:00	Supply moves; both curves move
11:30	Nine cases, and a teaser
11:38	Before Thursday 15 October

Slide 2 Plan for today

The class at a glance. First a last look at Monday's rule; then a real price that more than doubled, and the question of which curve moved. The crossing of the two curves, what happens above and below it, the price as a signal, and the three steps for a curve that moves. After the break supply moves, both curves at once, nine cases and a teaser for next week.

Two halves of fifty minutes with a ten-minute break between them.

Monday, one more look: milk gets dearer — which curve moves?

The neighbourhood's market for càfè amb llet. Milk goes up in price.

A) Demand: a dearer cup, fewer buyers

B) Supply: an input got dearer ✓

C) Both curves

D) Neither: nobody changed the price of a cup

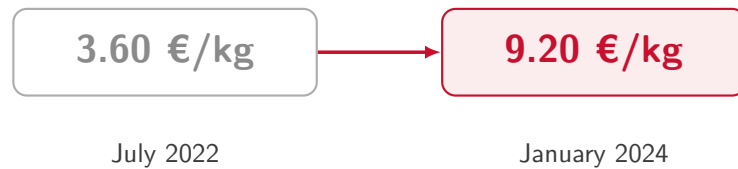
Milk is the bars' cost, not the buyers'. Demand asks what buyers want at each price; nothing about them changed. The price of a cup will change — today's story — as a result, not a cause.

Slide 3 Monday, one more look: milk gets dearer, which curve moves?

Monday's example once more: milk gets dearer in the neighbourhood's market for càfè amb llet. Milk is a cost of making a cup, so the event touches the sellers' side: at every price the bars offer fewer cups, and supply shifts (B). Demand asks what buyers want at each price, and nothing about the buyers changed: not their income, not their tastes, not the price of croissants.

A is the trap: the cup will get dearer, but that is the result we find today, not the event. D is the other trap: nobody changed the price of a cup, which is exactly why a curve moved instead of a point sliding along it. Which curve? The side the event touches: step 1 of everything today.

Olive oil at origin: 3.60 euros a kilo, then 9.20



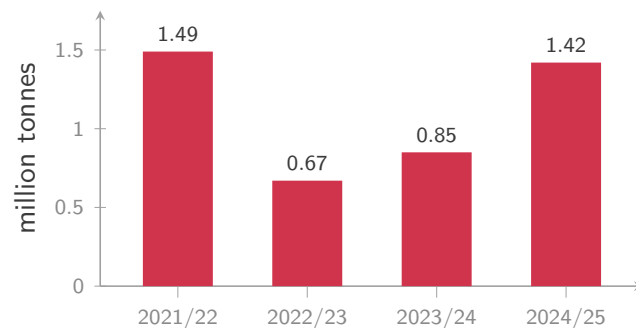
What happened — and which curve?

Poolred, the Spanish olive-oil origin price system: weekly prices paid to mills for extra virgin oil.

Slide 4 Olive oil at origin: 3.60 euros a kilo, then 9.20

The hook. The price at origin is what the mills get, before bottling and shops. Extra virgin olive oil cost 3.60 euros a kilo in July 2022 and 9.20 in January 2024, the highest ever recorded: two and a half times the price in eighteen months. What happened, and, in Monday's words, which curve? The next slide answers.

The 2022/23 harvest was less than half the one before



Two dry years; then the harvest came back, and the price came down: 4.25 by the end of 2024.

Olive oil made in Spain, by campaign: MAPA campaign balances. Prices: Poolred, extra virgin at origin, weekly.

Slide 5 The 2022/23 harvest was less than half the one before

The reveal. The bars are the olive oil made in Spain by campaign, which runs from the autumn harvest to the next summer: 1.49 million tonnes in 2021/22; 0.67 in 2022/23, after a dry and very hot year, less than half; 0.85 in 2023/24, still short; 1.42 in 2024/25, when the rain came back. The oil came back and the price came down: 4.25 euros a kilo by the end of 2024.

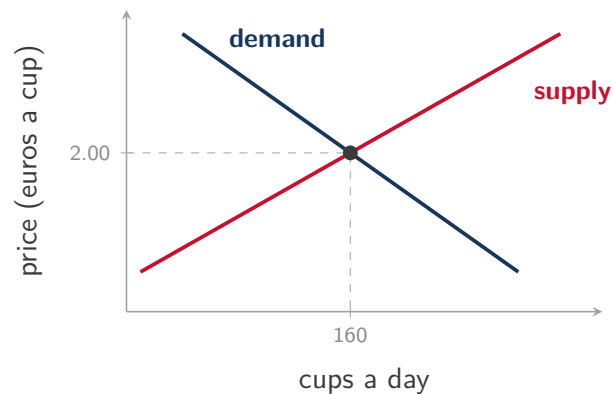
So a sellers' story, the harvest, and a price that went up and came down again. After the break the method takes it apart; for now it is the picture of a market.

Where the two curves cross: 160 cups a day at 2.00 euros

price	wanted	offered
1.00	240	60
1.50	200	110
2.00	160	160
2.50	120	210
3.00	80	260

the neighbourhood, cups a day

Equilibrium price: the one price at which wanted = offered. No queue, nothing left over.



Slide 6 Where the two curves cross: 160 cups a day at 2.00 euros

Monday's last picture, now with numbers: the whole neighbourhood's buyers (navy) and sellers (red) of càfé amb llet, in cups a day. The table lists what buyers want and what sellers offer at each price, and only one row agrees: at 2.00, 160 cups wanted and 160 offered. That price is the equilibrium price, or market price, and 160 cups the equilibrium quantity.

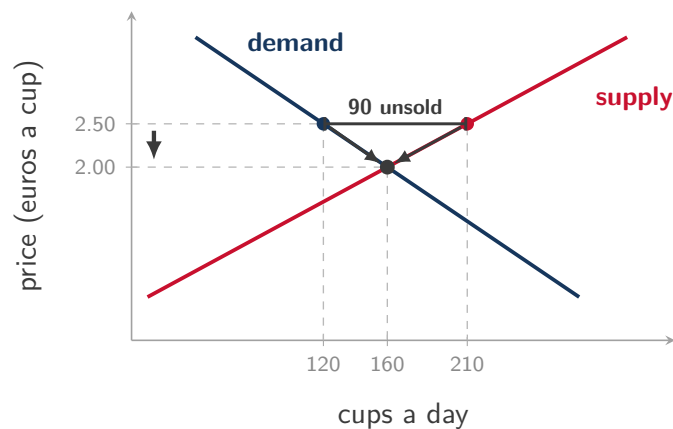
At 2.00 every buyer who wants a cup at that price gets one and every seller who wants to sell at that price sells: no queue, nothing left over. The market clears. The word is borrowed from physics, a state of rest with no force pushing; the next two slides are the forces.

Slide 7 At 2.50: offered 210, wanted 120, a surplus of 90

Suppose the bars try 2.50. Buyers want 120 cups, sellers offer 210: 90 cups a day unsold. That is a surplus, or excess supply: milk poured away, staff standing around. A seller with unsold cups cuts the price; as it falls, buyers come back down their curve and sellers make fewer along theirs, and the gap closes at 2.00.

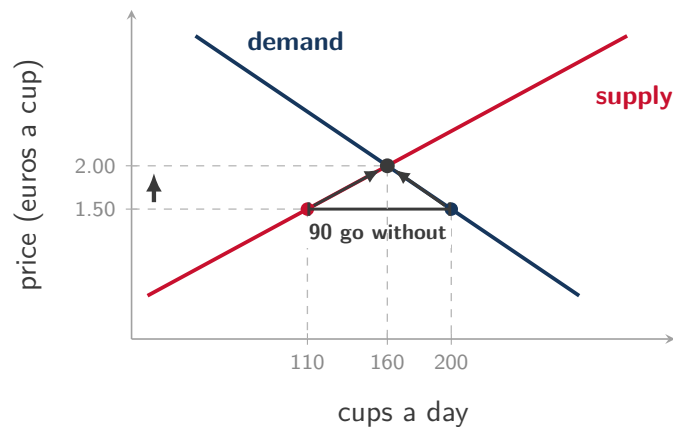
The two small arrows run along the curves. Nothing shifted: the price moved, and both sides slid along their own curve.

At 2.50: offered 210, wanted 120 — a surplus of 90



Surplus: unsold cups. Sellers cut the price; down both curves to 2.00.

At 1.50: wanted 200, offered 110 — a shortage of 90



Shortage: a queue. Sellers raise the price; up both curves to 2.00.

Slide 8 At 1.50: wanted 200, offered 110, a shortage of 90

The mirror image at 1.50. Buyers want 200 cups, sellers offer 110: a shortage, or excess demand, of 90: a queue, people turned away, the sold-out sign. A seller with a queue raises the price without losing sales; some buyers drop out up their curve, sellers make more up theirs, and the market is back at 2.00.

The reading calls this the law of supply and demand: the price adjusts until the quantity supplied and the quantity demanded are equal. Comparing one resting point with another is what the book calls comparative statics, the method of the second hour.

In pairs, three minutes: bikinis at the bars around the faculty

Where does the market settle? What happens at 5.50, and at 4.00?

price (€)	wanted	offered
4.00	300	80
4.50	250	140
5.00	200	200
5.50	150	260
6.00	100	320

bikinis a day

- 1 The crossing: **5.00 and 200 bikinis a day: wanted = offered.**
- 2 At 5.50: **a surplus of 110 (260 offered, 150 wanted); the price falls.**
- 3 At 4.00: **a shortage of 220 (300 wanted, 80 offered); the price rises.**

Slide 9 In pairs: bikinis at the bars around the faculty

Exercise 1: the bikini, the toastie sold at the bars around the faculty, in bikinis a day. The crossing is 5.00 and 200 a day, the only row where wanted equals offered. At 5.50 the bars offer 260 and students want 150: a surplus of 110, and the price falls. At 4.00 students want 300 and the bars offer 80: a shortage of 220, and the price rises.

Two traps. A surplus does not mean the sellers are happy: it is stock they cannot sell. And at 4.00 the market does not sell 300 bikinis but 80, the short side: you cannot buy what is not offered.

Nobody counts the cups. The price does.

The price rises

buyers: fewer cups, or tea
sellers: more cups, longer hours

The price falls

buyers: more cups
sellers: fewer cups, shorter hours

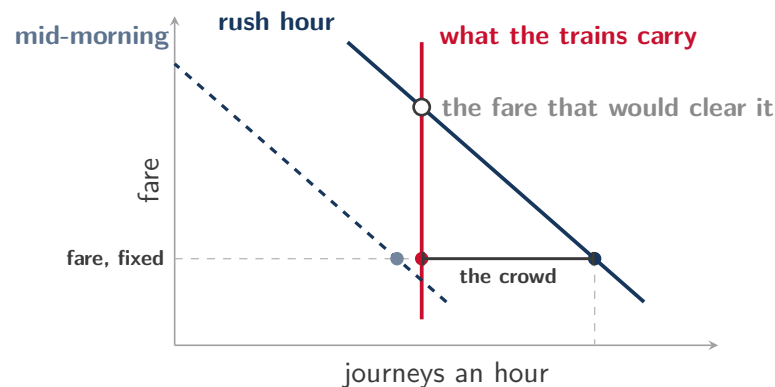
Prices as signals: one number tells buyers what to give up and sellers what it pays to make.
160 cups a day, planned by nobody.

Slide 10 Nobody counts the cups. The price does.

No office decided on 160 cups a day. The price did it, and it does two jobs at once. To buyers it says what a cup costs in other things given up: at 2.50 some decide a tea or a walk is worth more. To sellers it says what it pays to make: at 2.50, a second barista and longer hours. A rising price tells sellers to make more and buyers to give something up; a falling price says the reverse.

One number, seen by everyone, read by everyone, chosen by no one: a decentralised market. Whoever is willing and able to pay the price gets the good; whether that is fair is a normative question, and the model only says what happens.

When the price cannot move, the crowd does the rationing



A fare set by decree: the shortage is the full platform, not a price. Price controls: Class 9.

Slide 11 When the price cannot move, the crowd does the rationing

Monday's metro, where the price cannot move: the fare is set by the transport authority. The red line is what the trains can carry in an hour; the navy curve is the journeys people want at each fare at rush hour, the dashed one mid-morning. At the fixed fare, rush-hour demand runs past the capacity: the gap is the crowd on the platform, a shortage that cannot become a higher price, so it becomes waiting.

The hollow circle is the fare that would clear the platform; nobody proposes it. When the price is fixed, something else does the rationing: queues, crowding, first come first served. Price controls: Class 9.

When a curve moves

Three steps, and what the price does

slide 12 of 24

Slide 12 When a curve moves

The second half. Monday was the curves; now what happens to the crossing when one of them moves.

A change in the market: three steps, always the same

1

Which curve?

buyers' side: demand
sellers' side: supply
sometimes both

2

Which way?

right: more, left: less
at every price

3

The new crossing

draw it; read off
the price, the quantity

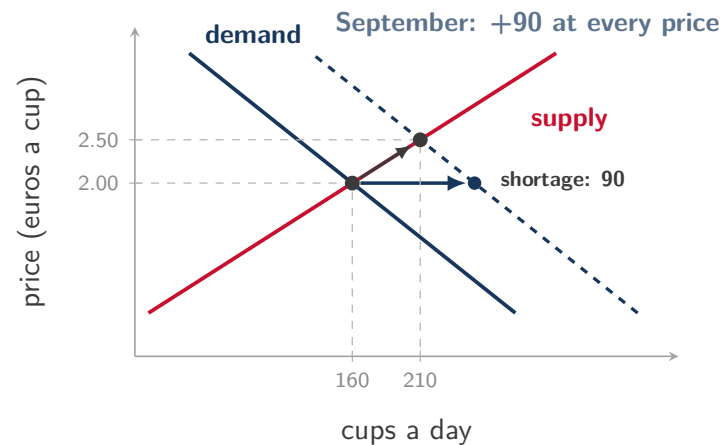
Step 1 is Monday's rule: anything but the good's own price moves a curve. The price is step 3.

Slide 13 A change in the market: three steps, always the same

The method for every question of this kind from now on. Step 1, which curve: the side the event touches, the buyers' (demand) or the sellers' (supply), sometimes both. Step 2, which way: right if more is wanted or offered at every price, left if less. Step 3: draw the new crossing and read off the new price and quantity.

Step 1 is Monday's rule: anything but the good's own price moves a curve. Step 3 is where the price comes in: it is the result, never the event. The common mistake is to skip step 3 and guess. Draw it.

Demand shifts right: a dearer cup, and more of them



Demand moved, supply did not: the market slid *up the supply curve* to 2.50 and 210 cups.

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14

slide 14 of 24

Slide 14 Demand shifts right: a dearer cup, and more of them

The coffee market at rest, 160 cups at 2.00, and an event: students are back in September, so there are more buyers. Step 1: demand. Step 2: right, the dashed curve, 90 more cups wanted at every price. At the old price buyers now want 250 and sellers offer 160, a shortage of 90, and slide 8 says what a shortage does: the price rises. Step 3: as it rises, sellers move up their supply curve and some of the new buyers drop out along the new demand curve; the new crossing is 2.50 and 210 cups.

The sentence to keep: demand moved, supply did not. The quantity supplied rose, along the same red line. Price and quantity both up.

Slide 15 Break

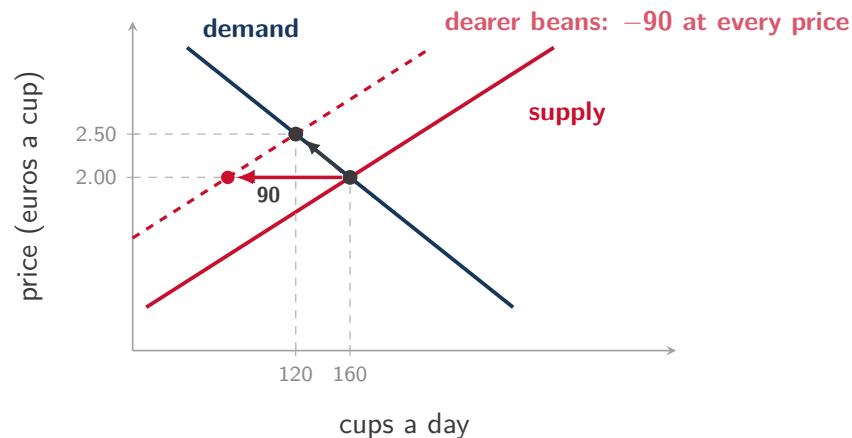
Ten minutes.

Break

Back at 11:00

slide 15 of 24

Supply shifts left: a dearer cup, and fewer of them



Supply moved, demand did not: the market slid *up the demand curve* to 2.50 and 120 cups.

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16

slide 16 of 24

Slide 16 Supply shifts left: a dearer cup, and fewer of them

The same market at rest and a second event: coffee beans get dearer, an input, from Monday's list of what moves supply. Step 1: supply. Step 2: left, 90 fewer cups offered at every price. At the old price sellers offer 70 and buyers want 160: a shortage of 90 again, so the price rises. Step 3: buyers move up their demand curve and buy fewer cups; the new crossing is 2.50 and 120 cups. Demand did not move.

Compare with slide 14: both events raised the price to 2.50, but one raised the quantity and the other lowered it. Which curve moved shows in the quantity, not in the price.

In pairs: the coffee market — which curve, which way, and then?

- 1 Milk gets cheaper.
supply right: the price falls, more cups
- 2 A heatwave: fewer people want a hot drink.
demand left: the price falls, fewer cups
- 3 Students are back in September.
demand right: the price rises, more cups
- 4 Baristas' wages rise.
supply left: the price rises, fewer cups
- 5 A cup gets dearer.
no curve moves by itself: a price is step 3's answer, never the event

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17

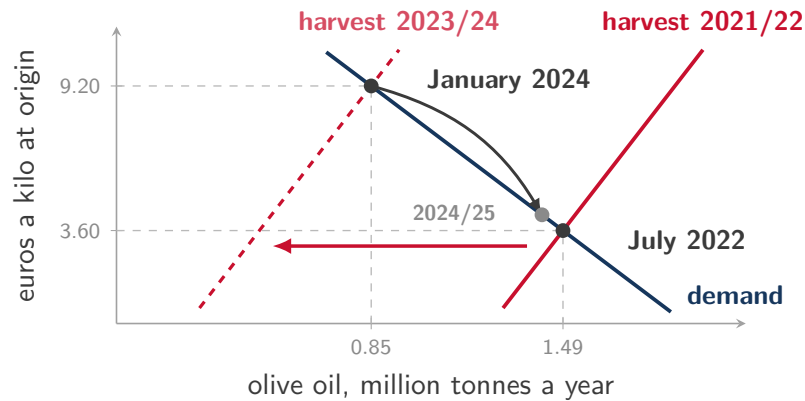
slide 17 of 24

Slide 17 In pairs: the coffee market, which curve, which way, and then?

Exercise 2: five events, steps 1 to 3 each time. 1 Milk gets cheaper: supply right; the price falls, more cups. 2 A heatwave: demand left; the price falls, fewer cups. 3 Students back in September: demand right; the price rises, more cups. 4 Baristas' wages rise: supply left; the price rises, fewer cups. 5 A cup gets dearer: no curve moves on its own; a price change is step 3's result, never the event.

Traps: answering 1 and 4 with demand, when costs are the sellers' side; and answering 5 with "the quantity demanded falls", true along a curve but no answer to what caused it. Note the pairing: 1 and 2 both lower the price, with opposite quantities.

Olive oil in three steps: supply left, up the demand curve, then back



Supply, then left, then up the demand curve: 3.60 to 9.20. Harvest back: 4.25 by late 2024.

Prices: Poolred, extra virgin at origin. Harvests: MAPA campaign balances. The curves are sketches, not data.

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18

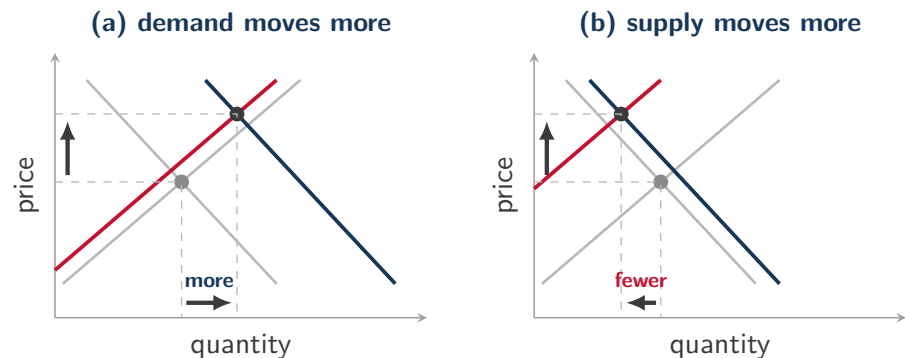
slide 18 of 24

Slide 18 Olive oil in three steps: supply left, up the demand curve, then back

The hook's payoff. Step 1: the harvest is the sellers' side: supply. Step 2: left, after two short harvests; the record price came in the second, so the dashed red line is the 2023/24 harvest. Step 3: up the demand curve to fewer kilos and a much higher price, 3.60 to 9.20: slide 16 with real numbers. The sequel: the 2024/25 harvest came back, supply shifted right again, and the market slid back down the demand curve to 4.25 by the end of 2024, the grey point.

Switching to sunflower oil at 9 euros is the move up the demand curve, not a shift of demand. The price rationed a small harvest without empty shelves: slide 10's signal at work.

Demand right, supply left: the price rises for sure; quantity depends



In grey, the curves before. Both shifts push the price up; they pull the quantity opposite ways.

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Introduction to Economics · Class 6

19

slide 19 of 24

Slide 19 Demand right, supply left: the price rises for sure; quantity depends

Both curves at once: demand right and supply left, slides 14 and 16 together, students back and dearer beans; the grey lines are the curves before. Both events push the price up, so the price rises for sure. The quantity is pulled both ways, up by the demand shift and down by the supply shift, and which wins depends on how far each curve moved. Panel (a), demand moved more: more cups. Panel (b), supply moved more: fewer.

So the honest answer for the quantity is that it depends on the sizes, which the model does not know and the data do. Writing that the price rises and the quantity could go either way is a full answer; writing that both rise is wrong.

In pairs: 23 April, the day of the roses

Everyone wants a rose; florists fly them in. The model says:

- A) Dearer roses, and more of them
- B) More roses for sure; the price: it depends ✓**
- C) Dearer roses for sure; how many: it depends
- D) Nothing for sure

Both curves right: each adds roses. Buyers push the price up, florists push it down; dearer only if buyers moved more than florists.

Slide 20 In pairs: 23 April, the day of the roses

Exercise 3: 23 April, when everyone in Barcelona wants a rose and florists fly them in from everywhere. Demand shifts right, massively; supply shifts right, massively. Both shifts add roses, so more roses for sure. On the price they disagree: buyers push it up, florists push it down, so roses are dearer only if buyers moved more than florists did. The answer is B.

A is the trap: it adds the price rise everyone remembers from their own 23 April, which the model does not promise. That roses are dearer on the day says that in Barcelona demand moves more than supply: a fact about the data, not a result of the model. C is slide 19's mirror: there the shifts agreed on the price; here on the quantity.

One curve moves: both known. Both move: one of them depends

	supply unchanged	supply right	supply left
demand unchanged	no change	$P \downarrow \cdot Q \uparrow$	$P \uparrow \cdot Q \downarrow$
demand right	$P \uparrow \cdot Q \uparrow$	$P ? \cdot Q \uparrow$	$P \uparrow \cdot Q ?$
demand left	$P \downarrow \cdot Q \downarrow$	$P \downarrow \cdot Q ?$	$P ? \cdot Q \downarrow$

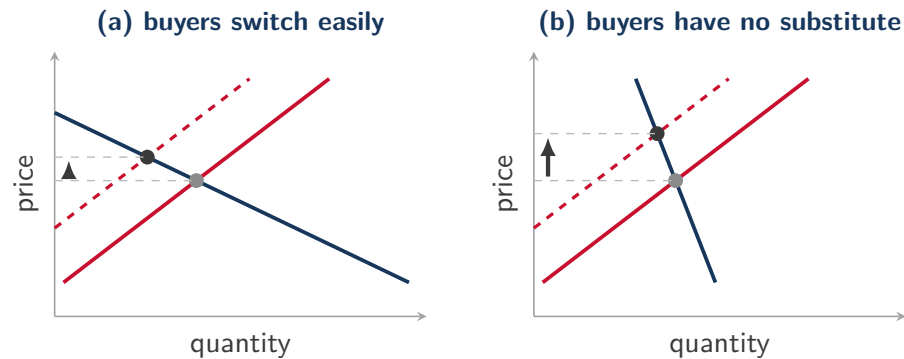
Do not memorise the table: draw the crossing. A ? is where the two shifts pull opposite ways.

Slide 21 One curve moves: both known. Both move: one of them depends

The nine cases in one table: the row is what demand did, the column what supply did. When one curve moves, the first row and the first column, both answers are known. When both move, exactly one of the two answers is a question mark: the one where the two shifts pull opposite ways. Demand right with supply right, the roses: more for sure, the price depends. Demand right with supply left, slide 19: dearer for sure, the quantity depends.

Do not memorise the table. Draw the crossing and the table writes itself.

The same shift, two markets: how far does the price jump?



Which way is today's question. How much is Thursday 15 October's: **elasticity**.

Slide 22 The same shift, two markets: how far does the price jump?

The teaser: the same supply shift to the left in two markets. In (a) buyers switch easily and the demand curve is flat: the price barely moves and the quantity falls a lot. In (b) buyers have no substitute and the demand curve is steep: the price jumps and the quantity hardly falls. Olive oil in a Spanish kitchen is nearer (b), which is why a smaller harvest meant such a big jump.

Today's question was which way. Next week's is how much: elasticity, the reading for Thursday 15 October.

Slide 23 In one line

The class in one line: the price moves until wanted equals offered; when a curve moves, find the new crossing. One curve moving: price and quantity both known. Both curves: one of the two depends on which moved more.

**The price moves until wanted equals offered.
A curve moves: find the new crossing.**

One curve: price and quantity both known.
Both curves: one of them depends on which moved more.

slide 23 of 24

Before Thursday 15 October

- No class Monday 12 October (Festa Nacional)
- Read Mankiw & Taylor, Ch. 3, pp. 53–66 — price elasticity of demand, other demand elasticities, price elasticity of supply; pp. 67–73 (applications) optional
- Project: your topic fixed by Thursday 15 October
- Problem set 1 (Blocks 1–2) handed out Thursday 15 October
- Practise in Sextant: the units the course page lists



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Slide 24 Before Thursday 15 October

No class on Monday 12 October; the next class is Thursday 15 October. Reading: Mankiw & Taylor chapter 3, pages 53 to 66, on the price elasticity of demand, the other demand elasticities and the price elasticity of supply; pages 67 to 73, the applications, are optional. Project: each group's question is fixed by Thursday 15 October, which is what today's meetings are for. Problem set 1, on Blocks 1 and 2, is handed out on Thursday 15 October. Practise in Sextant with the units the course page lists.

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